

Week 2

```
mysql> show databases;
+-----+
| Database |
+-----+
| bookflow_db |
| information_schema |
| librarydb |
| mysql |
| performance_schema |
| sakila |
| sql_hr |
| sql_inventory |
| sql_invoicing |
| sql_store |
| studentdb1 |
| sys |
| task_hub |
| userdb |
| world |
+-----+
15 rows in set (0.01 sec)
```

```
mysql> create database bookflow_db;
Query OK, 1 row affected (0.01 sec)
```

```
mysql> create table Books (
  -> bookd_id int primary key,
  -> title varchar(100) not null,
  -> isbn varchar(20) unique,
  -> published_year int check(published_year < 2027)
  -> );
Query OK, 0 rows affected (0.07 sec)
```

```
mysql> INSERT INTO Books (bookd_id, title, isbn, published_year) VALUES
    -> (1, 'The Great Gatsby', '9780743273565', 1925),
    -> (2, 'To Kill a Mockingbird', '9780061120084', 1960),
    -> (3, '1984', '9780451524935', 1949);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> select * from books;
+-----+-----+-----+-----+
| bookd_id | title                | isbn                | published_year |
+-----+-----+-----+-----+
| 1        | The Great Gatsby     | 9780743273565      | 1925           |
| 2        | To Kill a Mockingbird | 9780061120084      | 1960           |
| 3        | 1984                 | 9780451524935      | 1949           |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> create table members (
    -> member_id int primary key,
    -> full_name varchar(100),
    -> email varchar(100) unique
    -> );
Query OK, 0 rows affected (0.07 sec)
```

```
mysql> INSERT INTO Members (member_id, full_name, email) VALUES
    -> (101, 'John Smith', 'john.smith@email.com'),
    -> (102, 'Emma Wilson', 'emma.wilson@email.com'),
    -> (103, 'Michael Brown', 'michael.brown@email.com');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> select * from members;
+-----+-----+-----+
| member_id | full_name            | email                |
+-----+-----+-----+
| 101       | John Smith           | john.smith@email.com |
| 102       | Emma Wilson          | emma.wilson@email.com |
| 103       | Michael Brown        | michael.brown@email.com |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE Loans (
->     loan_id INT PRIMARY KEY,
->     member_id INT,
->     bookd_id INT,
->     loan_date DATE,
->     FOREIGN KEY (member_id) REFERENCES Members(member_id),
->     FOREIGN KEY (bookd_id) REFERENCES Books(bookd_id)
-> );
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> insert into Loans(loan_id, member_id, bookd_id, loan_date) VALUES
-> (1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),
-> (3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),
-> (5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),
-> (7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),
-> (9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

```
mysql> select * from Loans;
+-----+-----+-----+-----+
| loan_id | member_id | bookd_id | loan_date |
+-----+-----+-----+-----+
| 1 | 101 | 1 | 2025-01-05 |
| 2 | 102 | 2 | 2025-01-08 |
| 3 | 103 | 3 | 2025-01-10 |
| 4 | 101 | 2 | 2025-02-01 |
| 5 | 102 | 1 | 2025-02-05 |
| 6 | 103 | 2 | 2025-02-12 |
| 7 | 101 | 3 | 2025-03-01 |
| 8 | 102 | 3 | 2025-03-07 |
| 9 | 103 | 1 | 2025-03-15 |
| 10 | 101 | 1 | 2025-04-01 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

```
mysql> SELECT m.full_name AS Member_Name, b.title AS Book_Title
-> FROM Loans l
-> INNER JOIN Members m ON l.member_id = m.member_id
-> INNER JOIN Books b ON l.bookd_id = b.bookd_id;
```

Member_Name	Book_Title
John Smith	The Great Gatsby
John Smith	To Kill a Mockingbird
John Smith	1984
John Smith	The Great Gatsby
Emma Wilson	To Kill a Mockingbird
Emma Wilson	The Great Gatsby
Emma Wilson	1984
Michael Brown	1984
Michael Brown	To Kill a Mockingbird
Michael Brown	The Great Gatsby

10 rows in set (0.00 sec)

```
mysql> select published_year, count(bookd_id) as Total_books
-> FROM Books
-> GROUP BY published_year
-> ORDER BY published_year;
```

published_year	Total_books
1925	1
1949	1
1960	1

3 rows in set (0.01 sec)

```
mysql> create table Donation_History (
-> donation_id int primary key,
-> bookd_id int,
-> donor_name varchar(100),
-> donation_date date,
-> FOREIGN KEY (bookd_id) REFERENCES Books(bookd_id)
-> );
```

Query OK, 0 rows affected (0.06 sec)

```
mysql> start transaction;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> insert into Books (bookd_id, title, isbn, published_year)
      -> VALUES (4, 'Animal Farm', '9780451526342', 1945);
Query OK, 1 row affected (0.00 sec)
```

```
mysql> insert into Donation_History (donation_id, bookd_id, donor_name, donation_date)
      -> VALUES (1, 4, 'Raj Kumar', CURDATE());
Query OK, 1 row affected (0.02 sec)
```

```
mysql> commit;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> create index idx_books_isbn ON Books(isbn);
Query OK, 0 rows affected (0.09 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> select * from Books where isbn = '9780451524935';
+-----+-----+-----+-----+
| bookd_id | title | isbn          | published_year |
+-----+-----+-----+-----+
|          3 | 1984  | 9780451524935 |          1949 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```